Retaining walls

A **retaining wall** is a structure that holds or retains a material (example soil) behind it.

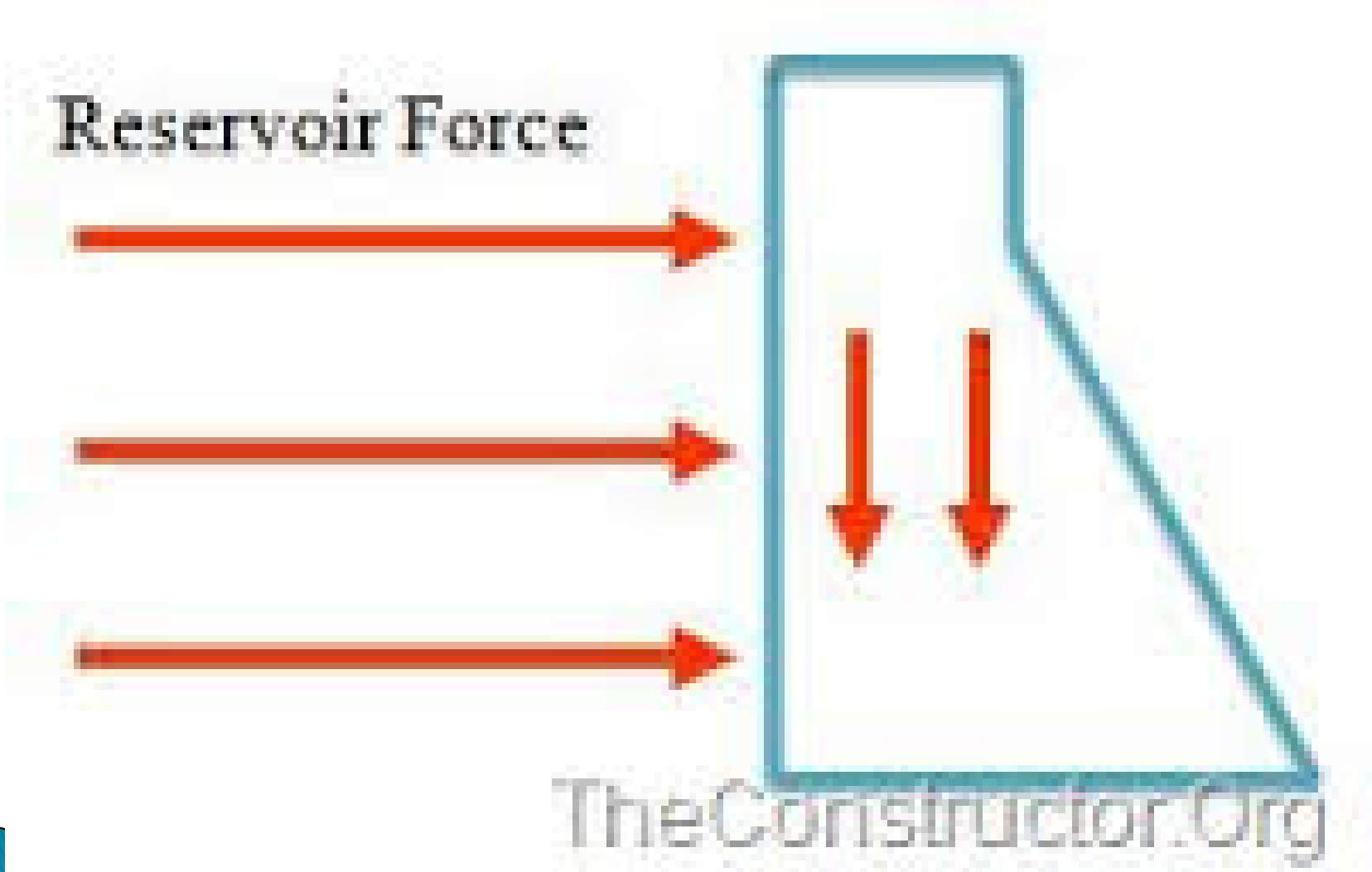
Purpose

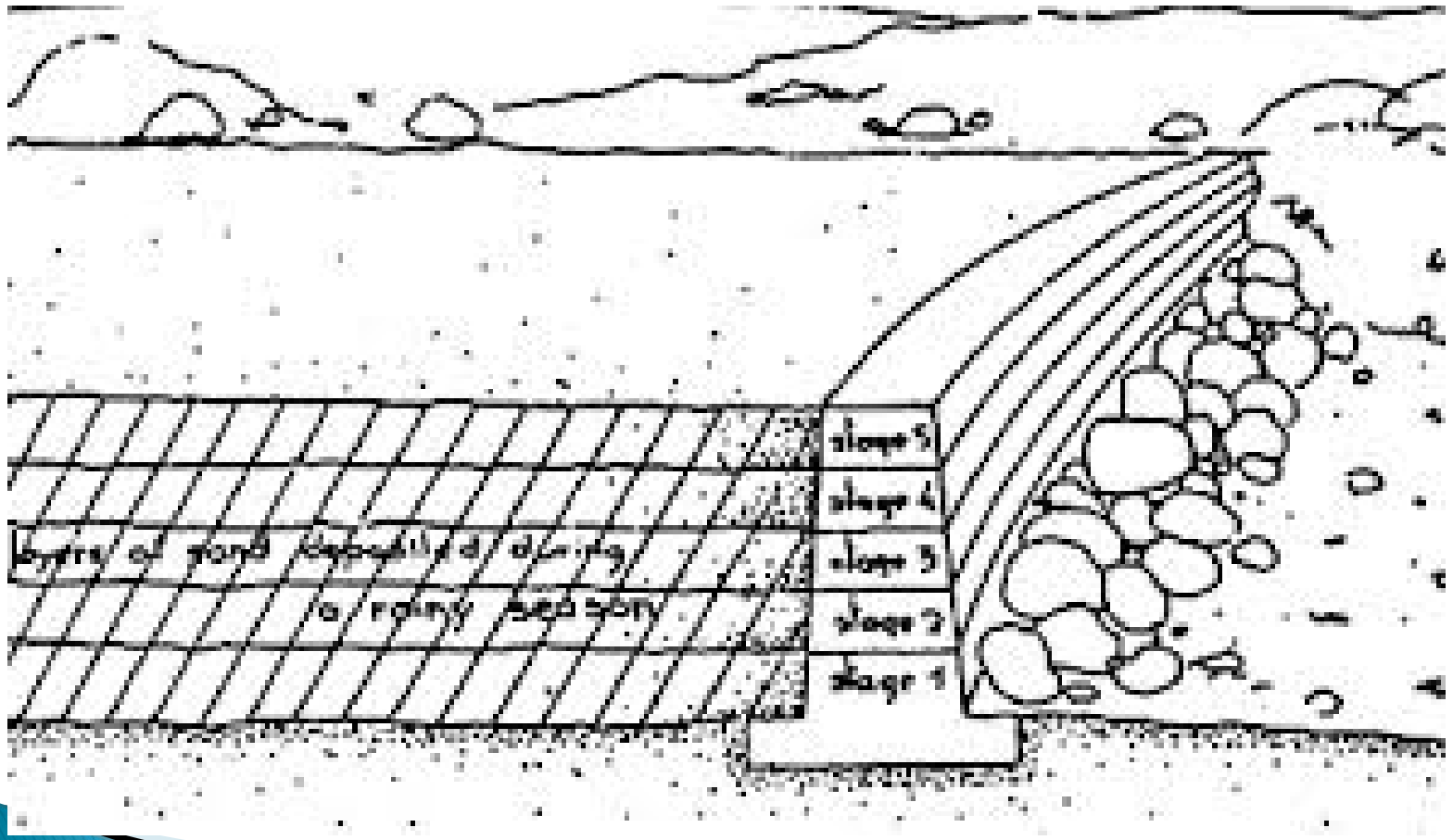
- Retaining Walls are commonly used to:
  - retain soil on sloping sites,
  - water in a pond or
  - bulk products within a storage area.
  - Eg. Dams

Retaining material walls

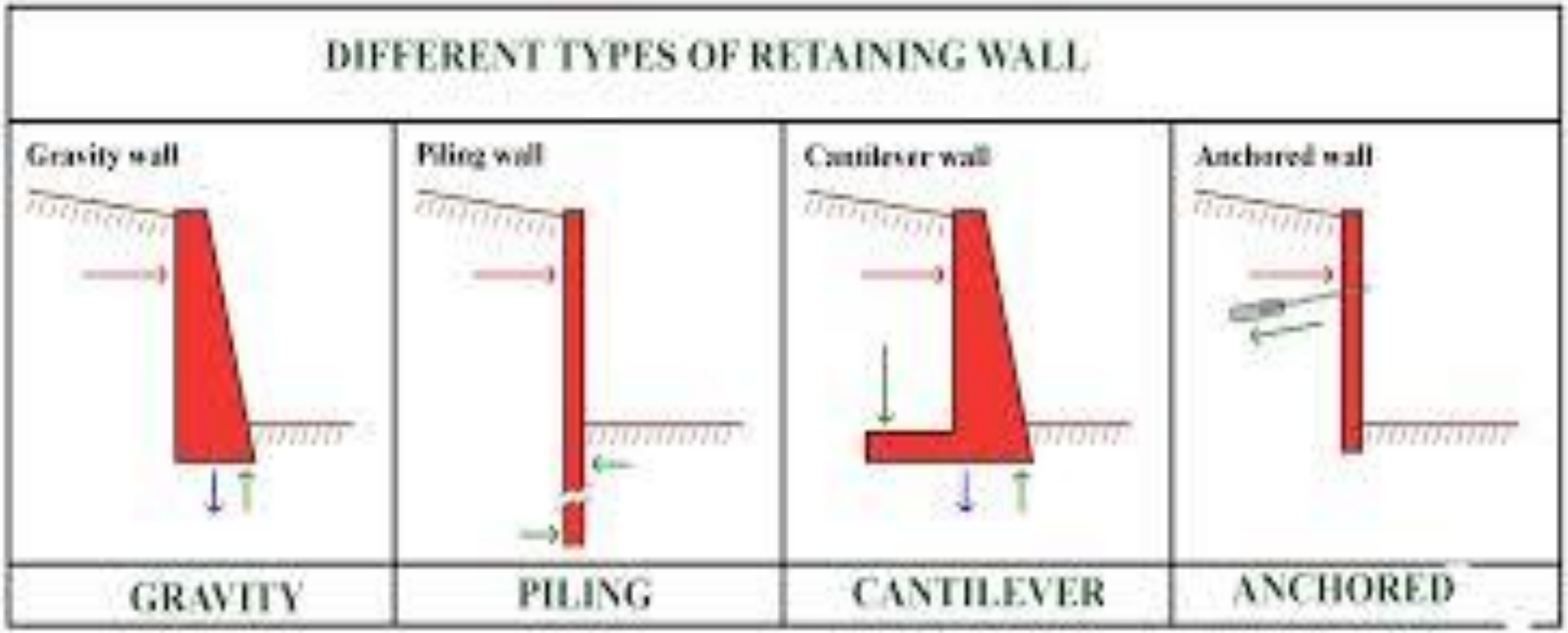
There are many types of materials that can be used to create **retaining walls**:

- concrete blocks,
- poured concrete,
- treated timbers,
- rocks or boulders.
- Some are easy to use, others have a shorter life span, but all can retain soil/feed/waste but not **water**.

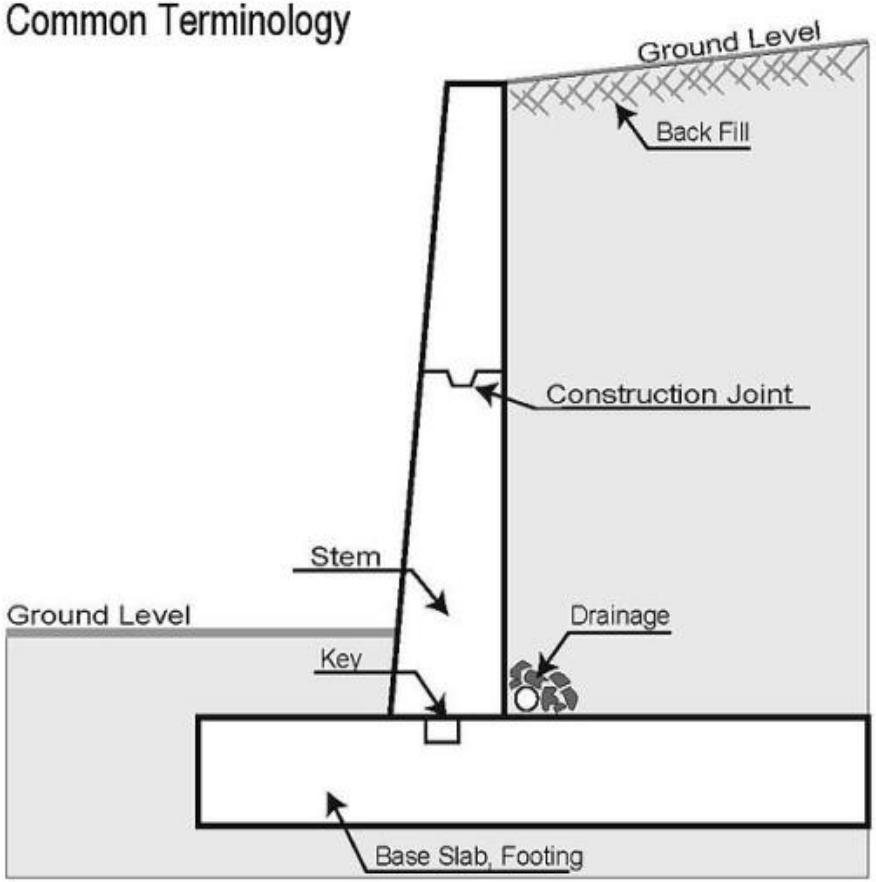

Reservoir Force
TheConstructor.Org


slope 5
slope 4
slope 3
slope 2
slope 1

Types


DIFFERENT TYPES OF RETAINING WALL
Gravity wall
Piling wall
Cantilever wall
Anchored wall
GRAVITY
PILING
CANTILEVER
ANCHORED

Common Terminology

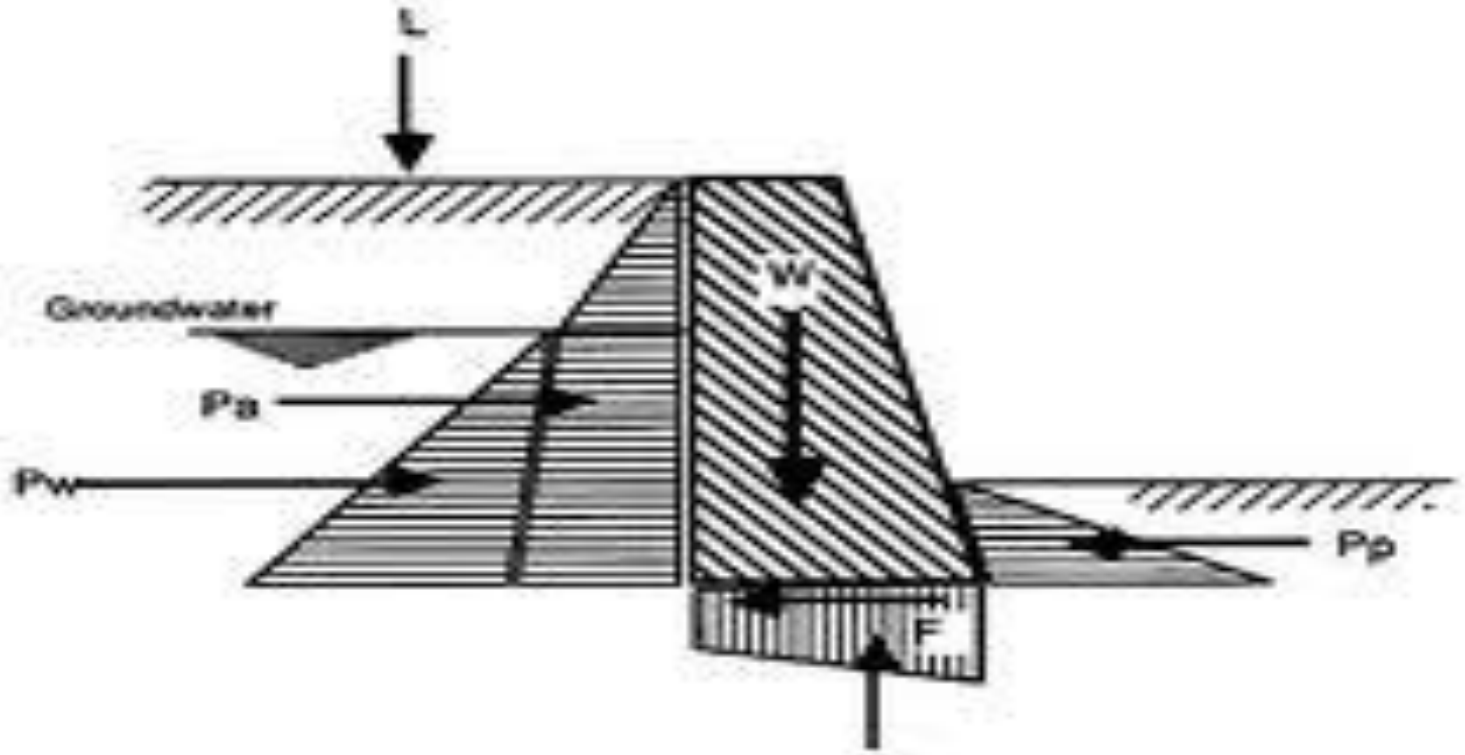


- There are several limiting conditions which, if exceeded, can lead to the failure of a retaining wall.

- Each must be addressed in designing a wall.

- The walls must resist the lateral pressures generated by either soils or in some cases water pressure.

FORCES ON RETAINING STRUCTURES



RESISTING

WATER PRESSURE (Pw)
PASSIVE SOIL PRESSURE (Pp)
SLIDING FRICTION (F)
FOUNDATION PRESSURE (Pf)

DRIVING FORCES

WATER PRESSURE (Pw)
ACTIVE SOIL PRESSURE (Pa)
WEIGHT (W)
SURCHARGE LOAD (L)

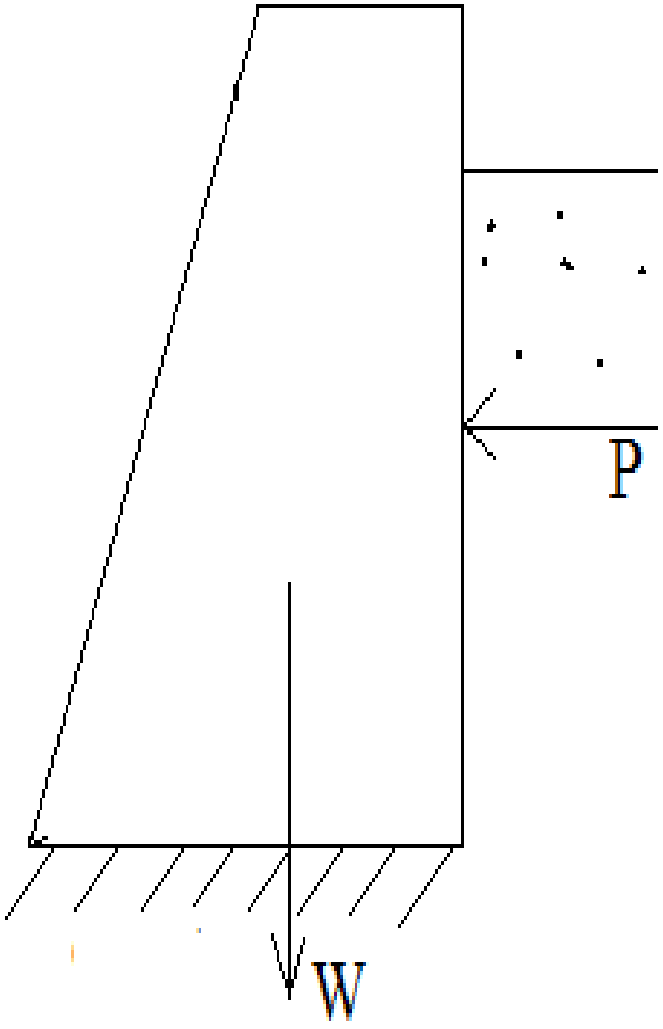
Wall failure

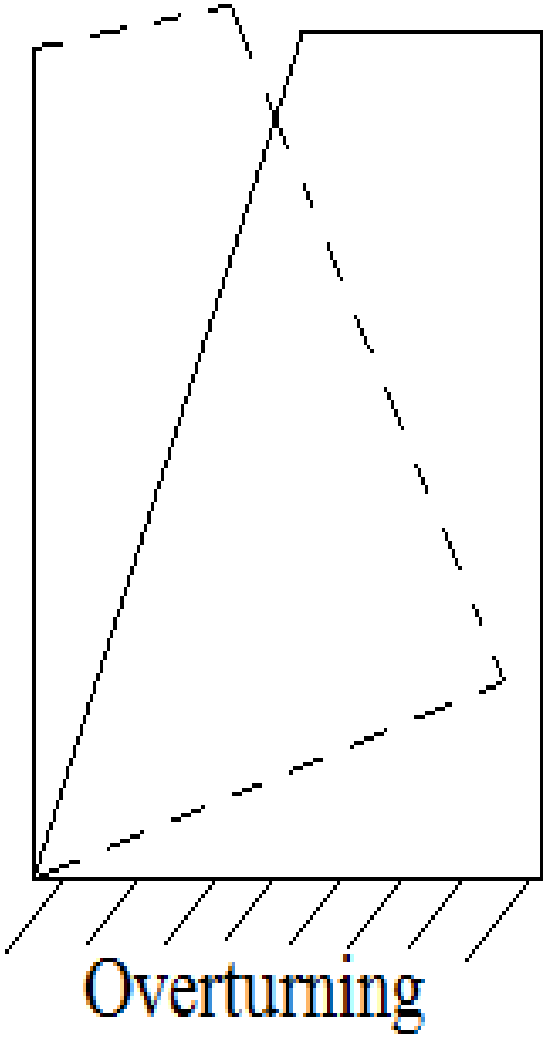
Wall failure can be as a result of:

- Overturning,
- Sliding,
- Bearing on ground,
- Rotational slip and
- Material failure

Overturning

- This occurs when the turning moment due to lateral forces exceeds that due to the self-weight of the wall.
- The factor of safety against overturning should be at least 2.





h- is the height at which force P acts

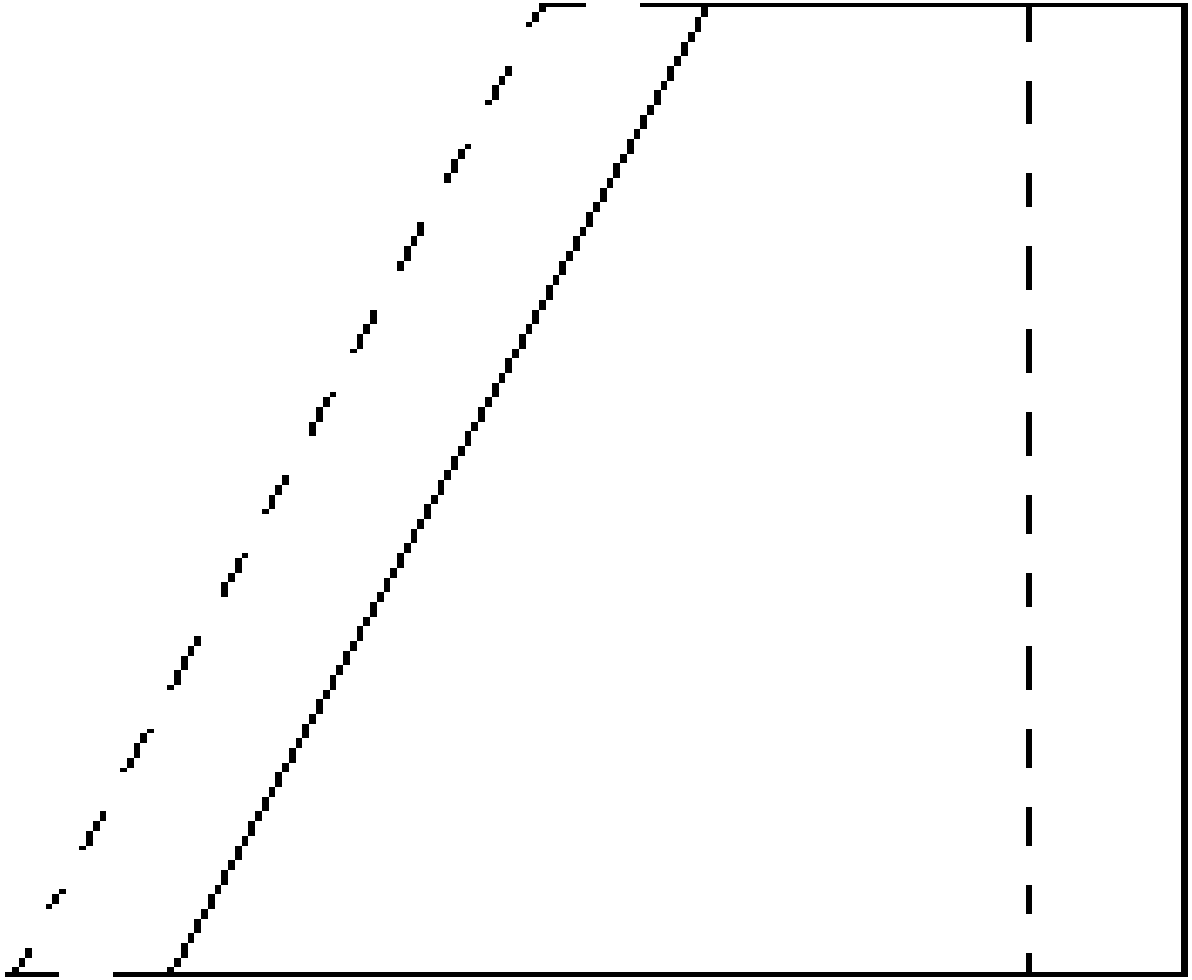
x- distance from edge to where W is acting

- Overturning moment = Stabilizing/restoring moment
- M_o= Ph
- M_s= Wx
- Factor of safety (F.S) is the ratio restoring moment/overturning moment =Wx/Ph

- h- is the height at which force P acts
- x- distance from edge to where W is acting

Sliding

- The wall will slide if the lateral thrust exceeds the frictional resistance developed between the base of the wall and the soil.
- The factor of safety against sliding should be about 1.8



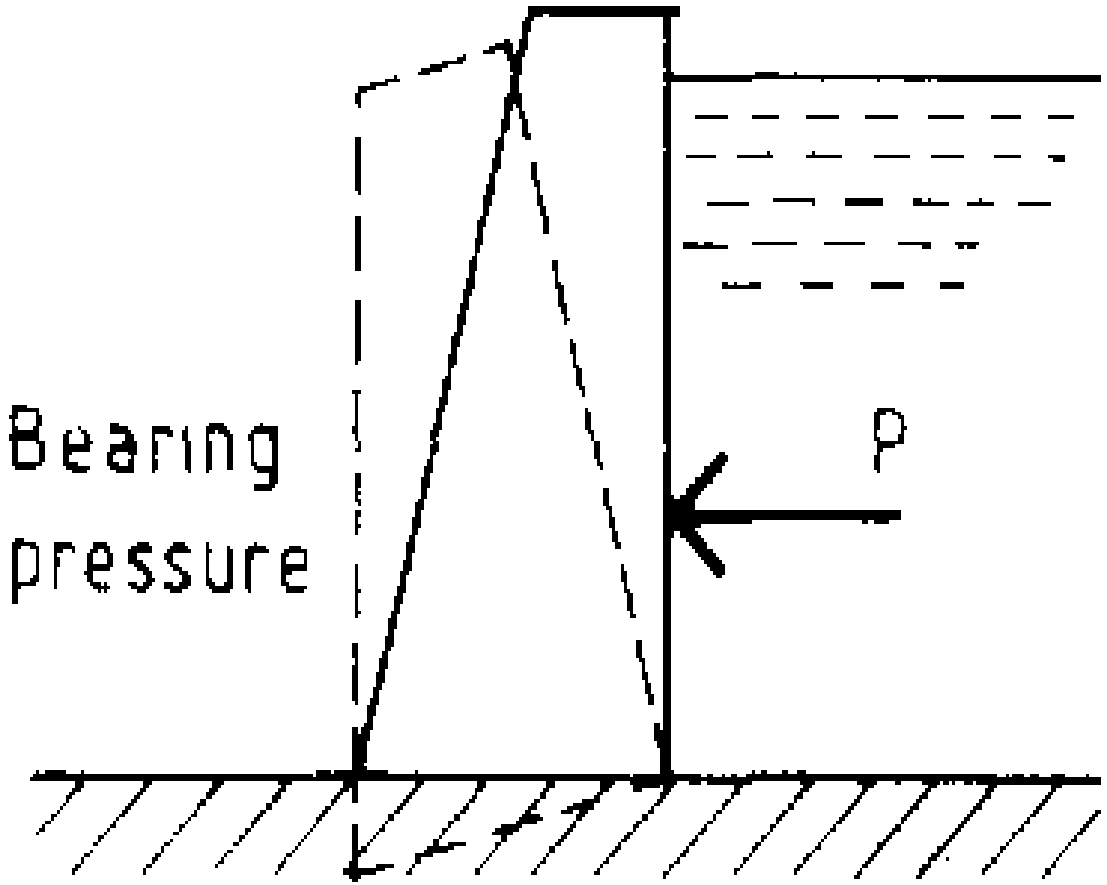
Sliding

- Sliding force =Resistive force
- Sliding force = P
- Resistive force = uW
- F.S= uW/P

Bearing on Ground

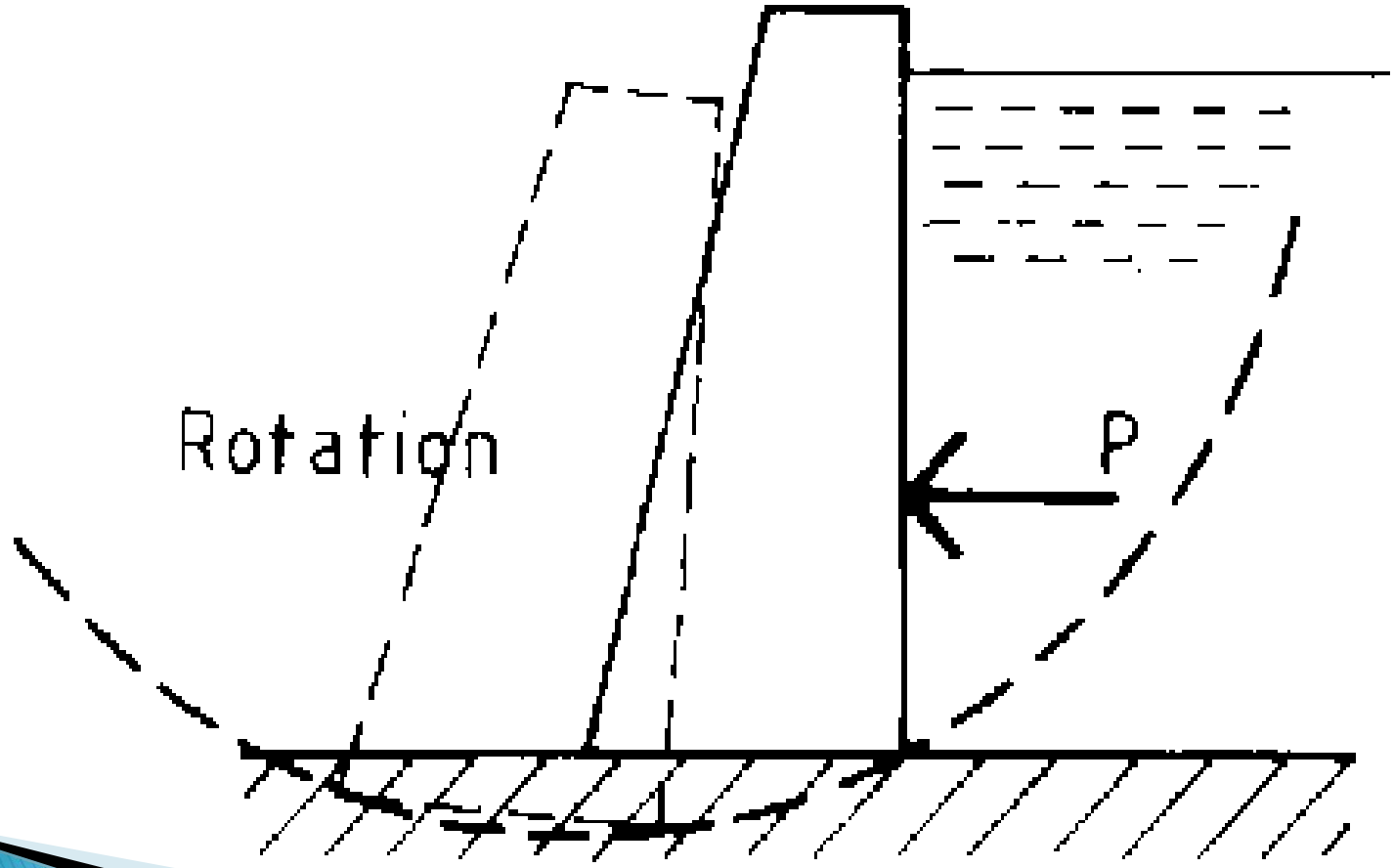
The normal pressure between the base of the wall and the soil beneath can cause a bearing failure of the soil, if the ultimate bearing capacity is exceeded.

- Usually the allowable bearing pressure will be one-third of the ultimate value. Note that the pressure distribution across the base is not constant.


Bearing
pressure
P

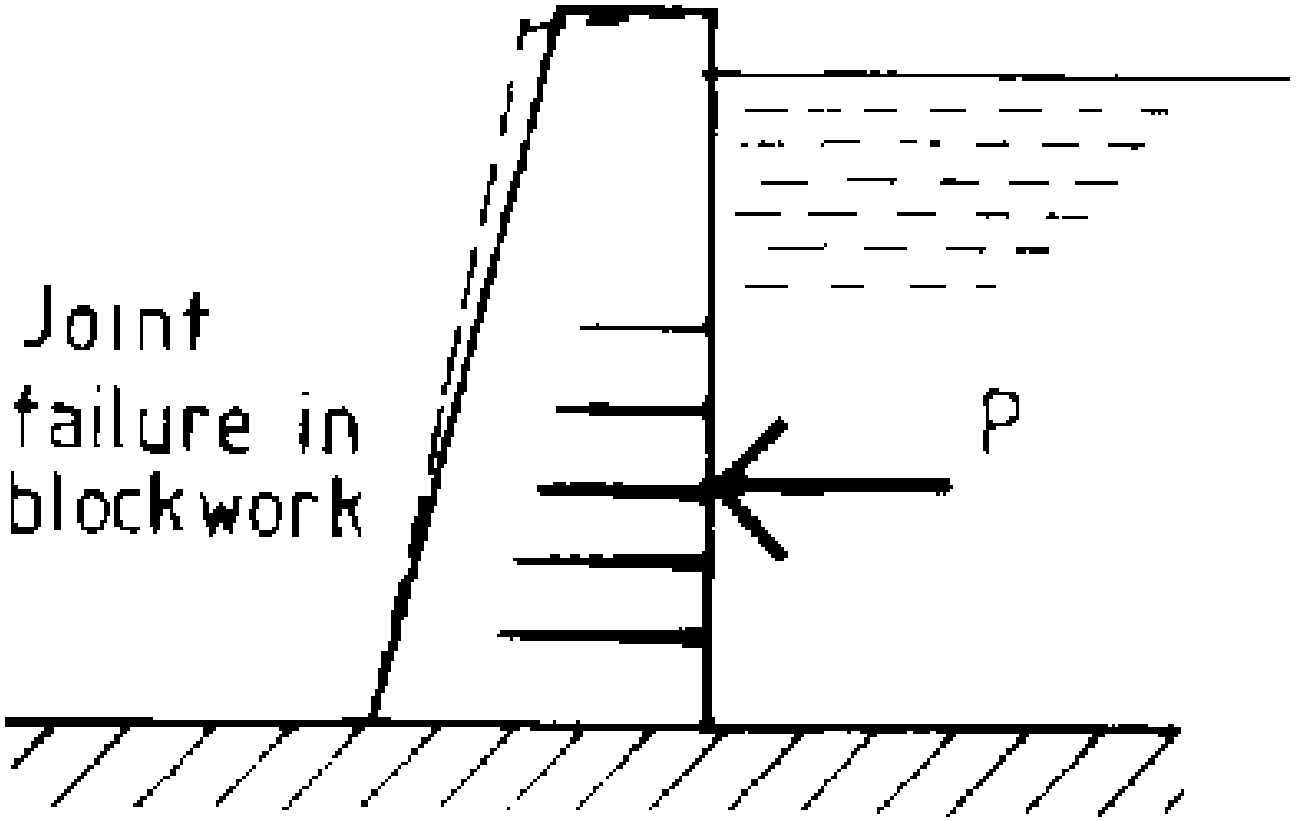
Rotational Slip -

- The wall and a large amount of the retained material rotate about some point O if the shear resistance developed along a circular arc is exceeded.
- The analysis is too complex to include here

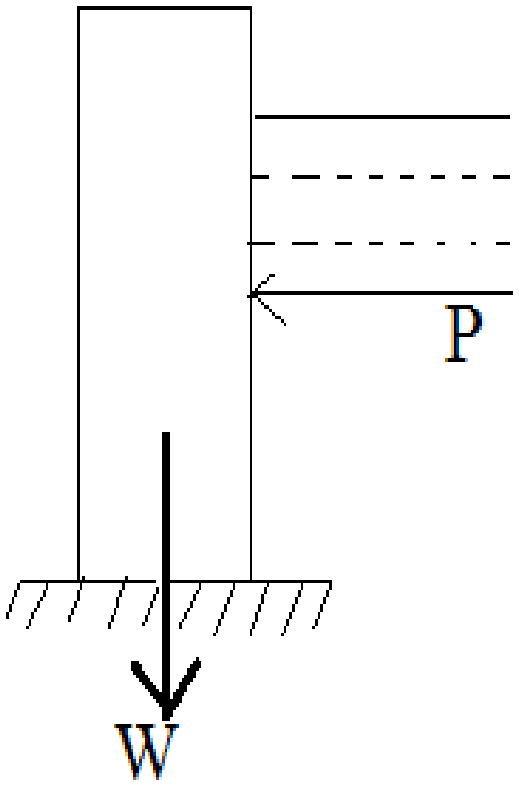

Rotation
P

Wall Material Failure -

- The structure itself must be capable of withstanding the internal stresses set up, that is, the stresses must not exceed allowable values. Factors of safety used here depend on the material and the level of the designer's knowledge in respect to the loads actually applied. Naturally, both shear and bending must be considered, but the most critical condition is likely to be tension failure of the 'front' facet


Joint
failure in
blockwork
P

- Analysis of retaining wall

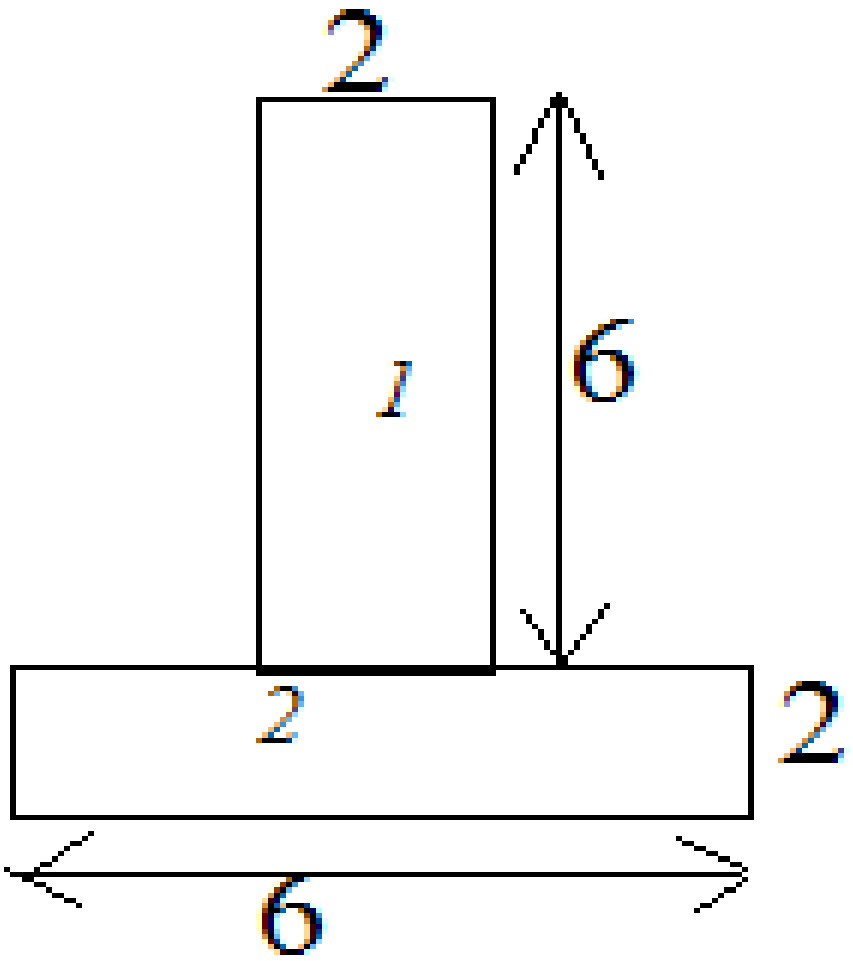


$$P = \rho g h \, (N/m^2)$$

$$H_2O \; Pressure: P = \frac{wH^2}{2} \, (N/m)$$

vertical compression of the base, W=vρg

Finding Centroid, which is the centre of a mass

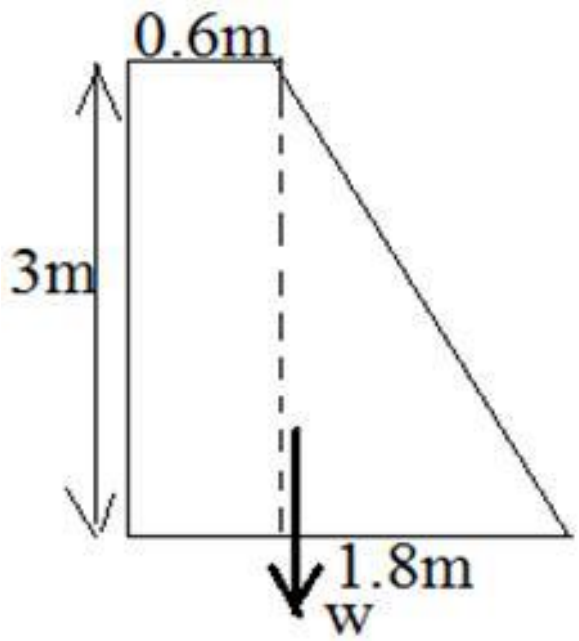


$$\bar{y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2}$$

$$\bar{y} = \frac{(6x2x3) + (6x2x3)}{(6x2) + (6x2)} = \frac{72}{24} = 3$$

A_1 and A_2 are the areas of 1 and 2

A concrete dam has a trapezoidal cross section as shown below. Water is impounded to a depth of 3m.



Assume coefficient of sliding friction, $\mu = 0.7$, $\rho_{H_2O} = 1000\ kg/m^3$, $\rho_{material} = 23\ kN/m^3$

- Find the resultant water force (horizontal trust i.e. force causing sliding)
- Force of 2m/1m length of wall
- Vertical compression force at the base
- Moment about centre line of the base
- Frictional resistance or the sliding resistance force
- Factor of safety for sliding